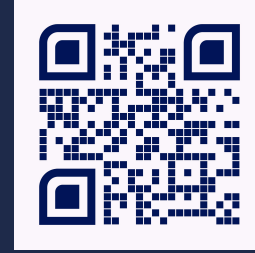


SEDTalker: Emotion-Aware 3D Facial Animation Using Frame-Level Speech Emotion Diarization

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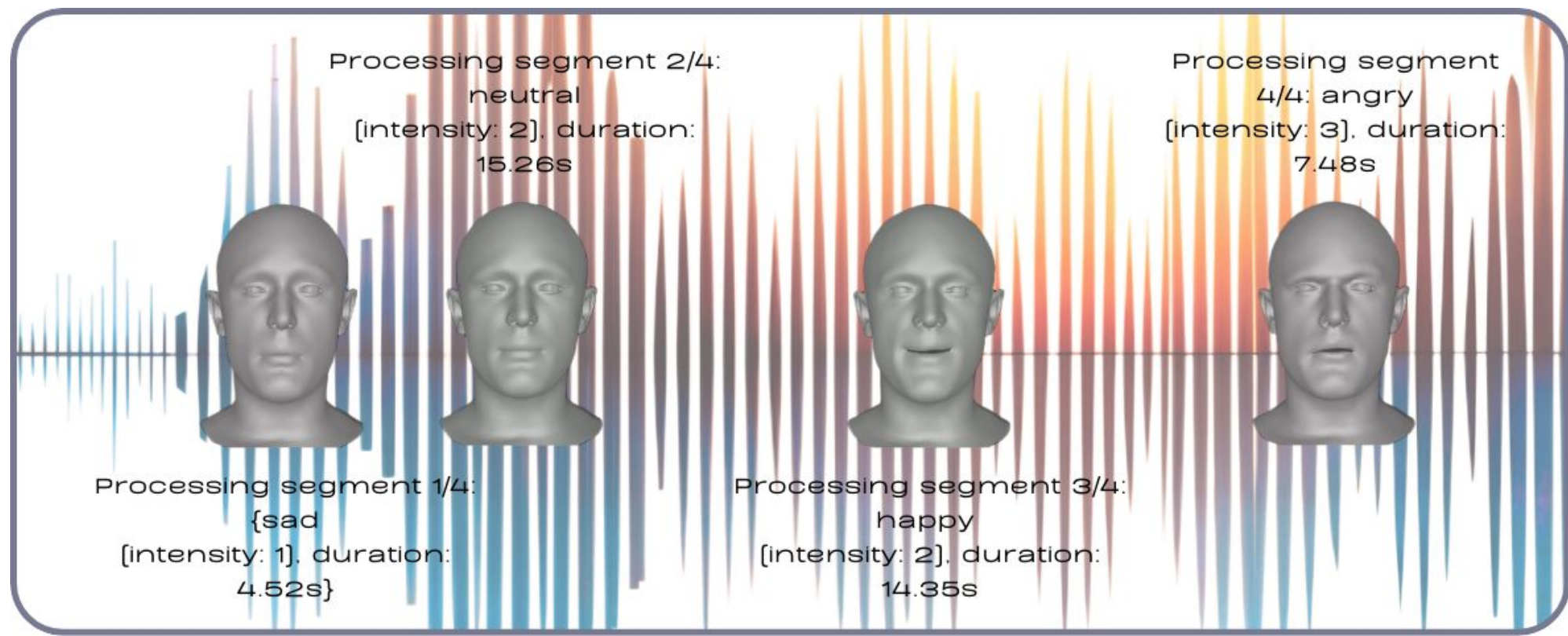


Fig. 1. **SEDTalker** is a speech-driven framework for emotion-aware 3D facial animation. By leveraging frame-level speech emotion diarization, the model captures fine-grained emotional cues and directly translates them into 3D facial parameters for precise expressive control.

MOTIVATION

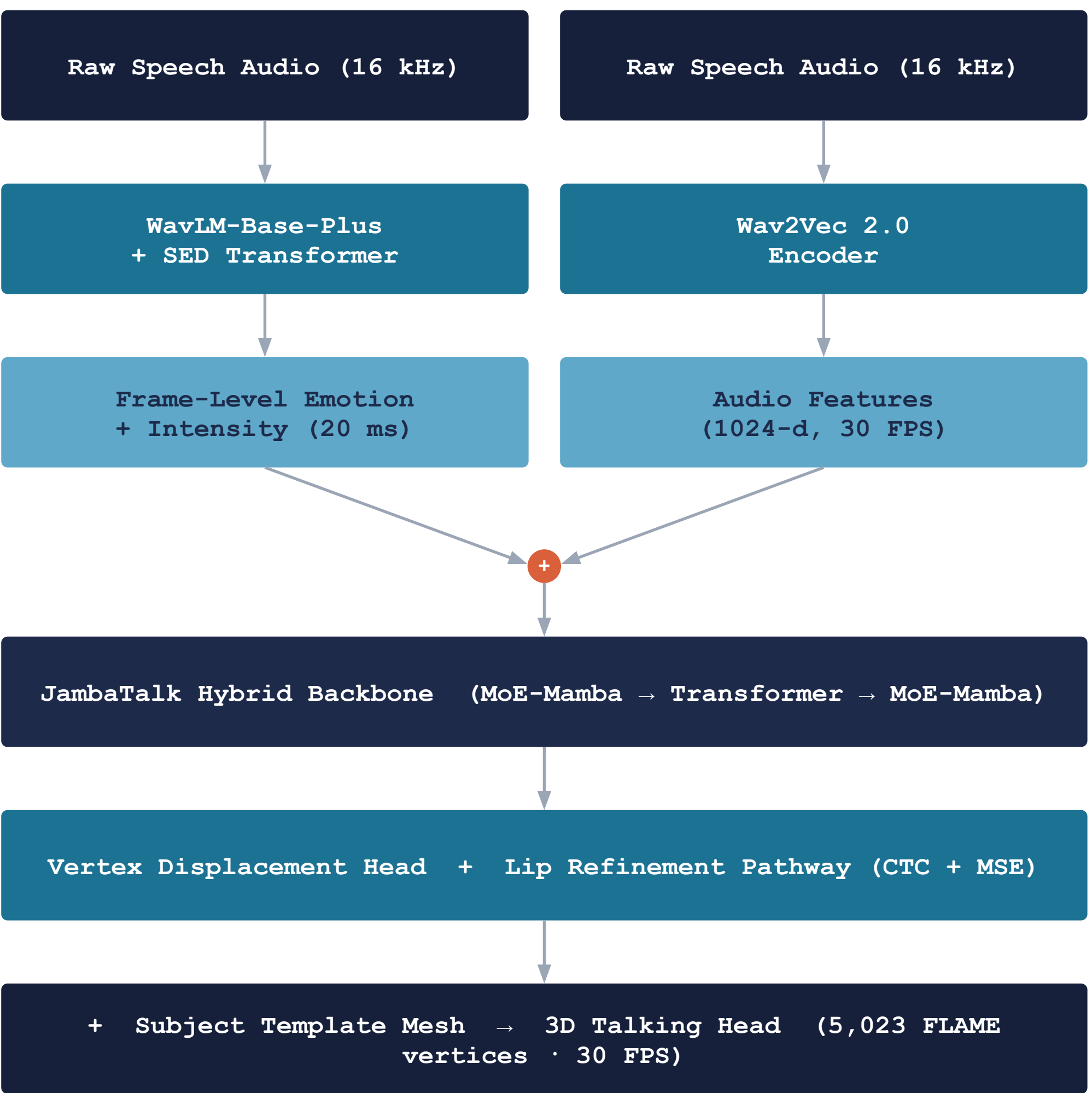
SEDTalker predicts emotion category and intensity frame-by-frame directly from speech, instead of relying on one manually set emotion label per utterance like prior work. Those dense emotion signals are turned into learned embeddings that condition a hybrid Transformer-Mamba 3D animation model, letting it separate speech content from emotional style while keeping identity and motion stable over time. Tested on a large multi-corpus speech-emotion-diarization dataset and on EmoVOCA for 3D facial animation, it achieves strong frame-level emotion recognition, low geometric/temporal error, and smooth, controllable emotion transitions, showing that frame-level diarization is an effective path to expressive, controllable talking heads.

KEY CONTRIBUTIONS

- Frame-level emotion-conditioned generation:** Formulates speech-driven 3D facial animation as a frame-level task, enabling continuous, intra-utterance emotion modulation that utterance-level approaches can't capture by design.
- Decoupled SED + animation training:** Separating speech emotion diarization from animation training removes the need for paired emotional-speech / 3D-mesh data, a bottleneck that blocked prior work.
- New frame-level emotion diarization corpus:** 58,834 utterances aggregated from nine public datasets, spanning 149 speakers and four English accents, achieving 78.9% weighted F1 at 20 ms resolution.
- Extended JambaTalk backbone:** Adds an additive emotion-intensity conditioning module that separates linguistic content from emotional style while preserving temporal coherence, running at just 13 GB GPU memory for 18 emotion-intensity states plus a neutral one.

METHOD OVERVIEW

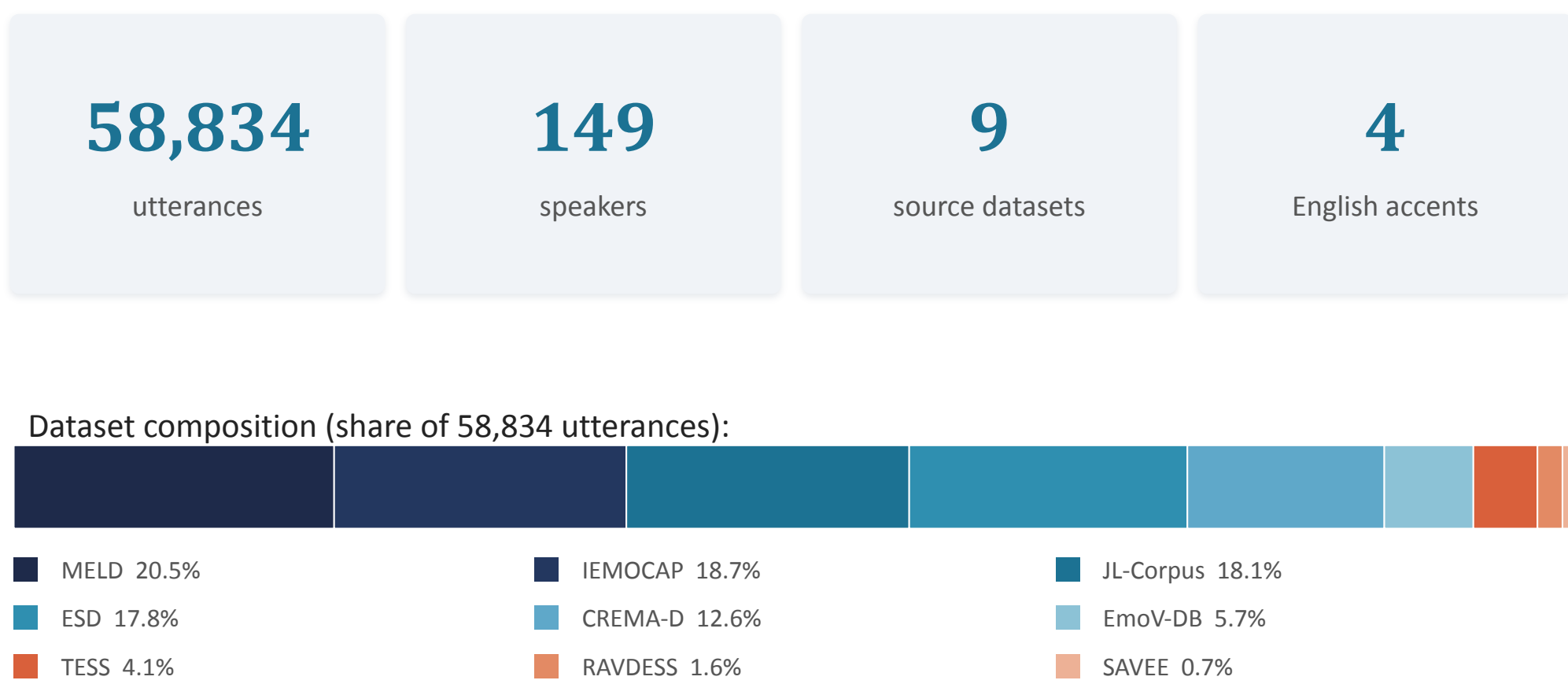
Two decoupled stages. (1) A Speech Emotion Diarization (SED) model predicts frame-level emotion category and intensity directly from raw audio. (2) A hybrid Transformer-Mamba animation backbone (JambaTalk) is conditioned on these dense signals to generate 3D vertex displacements at 30 FPS, with no need for paired emotional-speech / 3D-mesh training data.



WHY FRAME-LEVEL MATTERS

	Utterance-Level	Frame-Level (Ours)
Emotion granularity	One label / Clip	Dense (per 20 ms)
Intra-utterance transitions	Not modeled	Smooth & Continuous
Paired emotional 3D data	Often required	Not required
Expression realism	Flat / Abrupt	Fine-grained control

EMOTION DIARIZATION CORPUS



7-class taxonomy (Ekman's basics + an added "upset" class) with inverse-frequency loss weighting to counter severe class imbalance; rare classes like fear (weight 2.15) contribute more to the loss than dominant classes like happy (weight 0.31).

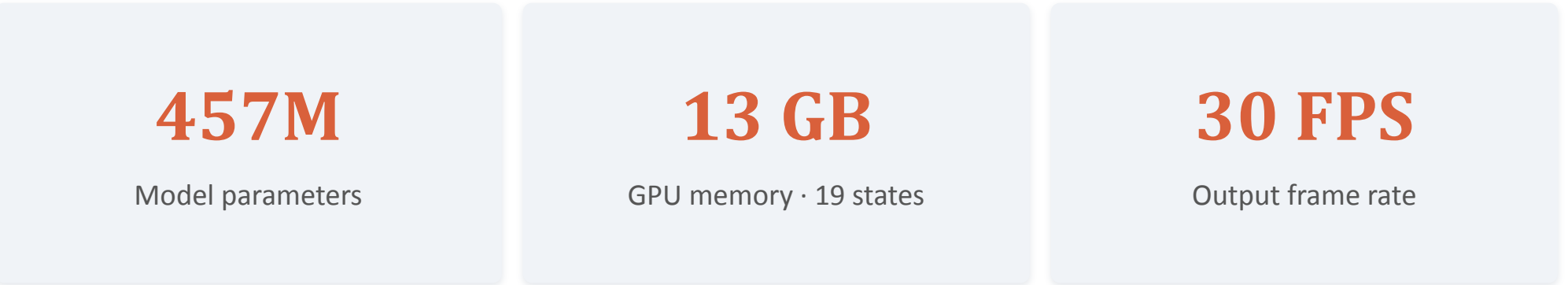
SPEECH EMOTION DIARIZATION MODEL

WavLM-Base-Plus (pre-trained on 94k hours of unlabeled speech) feeds a 12-layer Transformer encoder. Selective fine-tuning keeps the CNN feature extractor and the first 6 Transformer layers frozen, adapting only the upper 6 layers, cutting trainable parameters from 95M to 12.4M (13%). A frame-level MLP head (768→512→128→7) is trained with inverse-frequency-weighted cross-entropy.



3D ANIMATION BACKBONE

Emotion/intensity conditioning is additive: $C = \text{Emb}(e) + W_i \cdot i$, broadcast across time and summed with projected Wav2Vec 2.0 features. The JambaTalk backbone alternates MoE-Mamba layers (2 experts, top-2 routing, $O(T)$ linear complexity) with a Transformer layer for global context. A dedicated lip-refinement pathway supervises the 254 FLAME mouth vertices via CTC and MSE losses against phonetic features.



TRAINING CONFIGURATION

SED model: AdamW (lr 1e-4, weight decay 1e-4), ReduceLROnPlateau (factor 0.5, patience 3), gradient accumulation $\times 8$ (physical batch 4 $\rightarrow$ effective 32), mixed precision, 50 epochs, ~ 30 h on one RTX 4090 (24GB). **Animation model:** Adam (lr 1e-4), batch size 1, 50 epochs on EmoVOCA, optimizing a composite objective:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{vert}} + \lambda_2 \mathcal{L}_{\text{vel}} + \lambda_3 \mathcal{L}_{\text{lip}} + \lambda_4 \mathcal{L}_{\text{ctc}}$$

$\lambda_1 = \lambda_1 = 1000$ · $\lambda_2 = 1 \times 10^{-3}$ · $\lambda_3 = 1 \times 10^{-4}$
(vertex, velocity, lip-sync, and CTC text-alignment terms)

SED PERFORMANCE

Evaluated on a 7,007-utterance held-out test set (1.85M frames, 10.3 h) at 20 ms resolution with 100 ms temporal smoothing, 78.92% frame accuracy, 78.9% weighted F1.

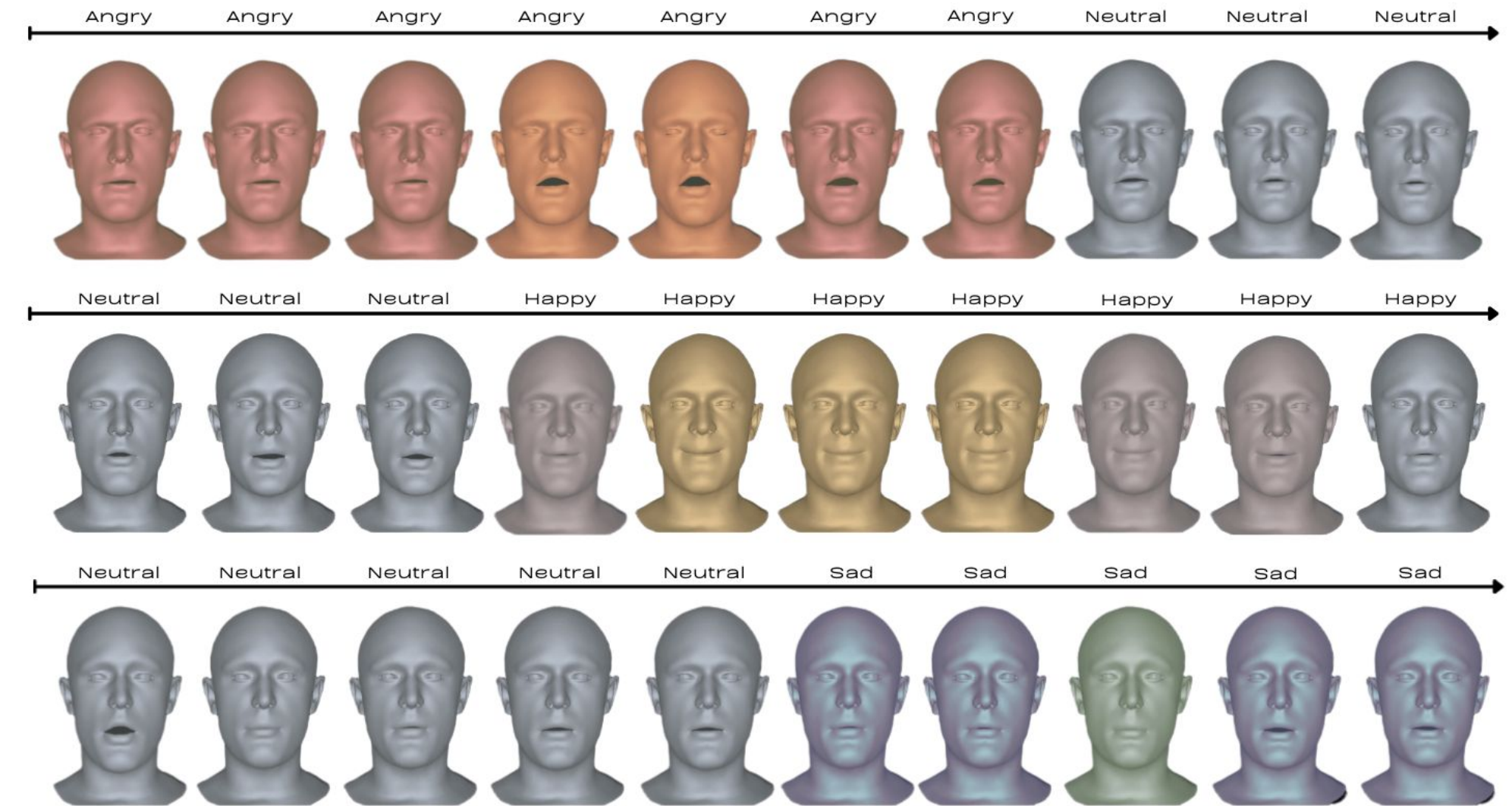


Fig. 2. Continuous emotion-controlled facial expression generation. The figure shows temporally ordered facial meshes generated along an emotion trajectory, transitioning across discrete emotion categories (angry, neutral, happy, and sad). Despite the discrete labels, facial geometry evolves smoothly with consistent identity preservation, demonstrating continuous interpolation and fine-grained control over expression intensity.

3D ANIMATION RESULTS

Across 5,544 test sequences (628,668 frames), reconstruction is accurate and stable: MVE 5.976×10^{-3} , LVE 7.847×10^{-3} , EVE 7.863×10^{-3} , FFE 1.848×10^{-3} , varying by less than $\pm 0.02 \times 10^{-3}$ across six emotions (afraid, disgusted, irritated, sad, pleased, and upset).

Method	MVE↓	LVE↓	EVE↓	FFE↓
FaceFormer	6.006	7.837	7.897	1.862
SEDTalker (Ours)	5.978	7.797	7.808	1.846
w/o Mamba	6.036	7.872	7.853	1.878
w/o Transformer	5.998	7.823	7.807	1.855

Quantitative comparison on EmoVOCA test set afraid emotion (939 train, 120 val, and 117 test). Both methods were trained on identical splits using the same evaluation protocol.

QUALITATIVE RESULTS

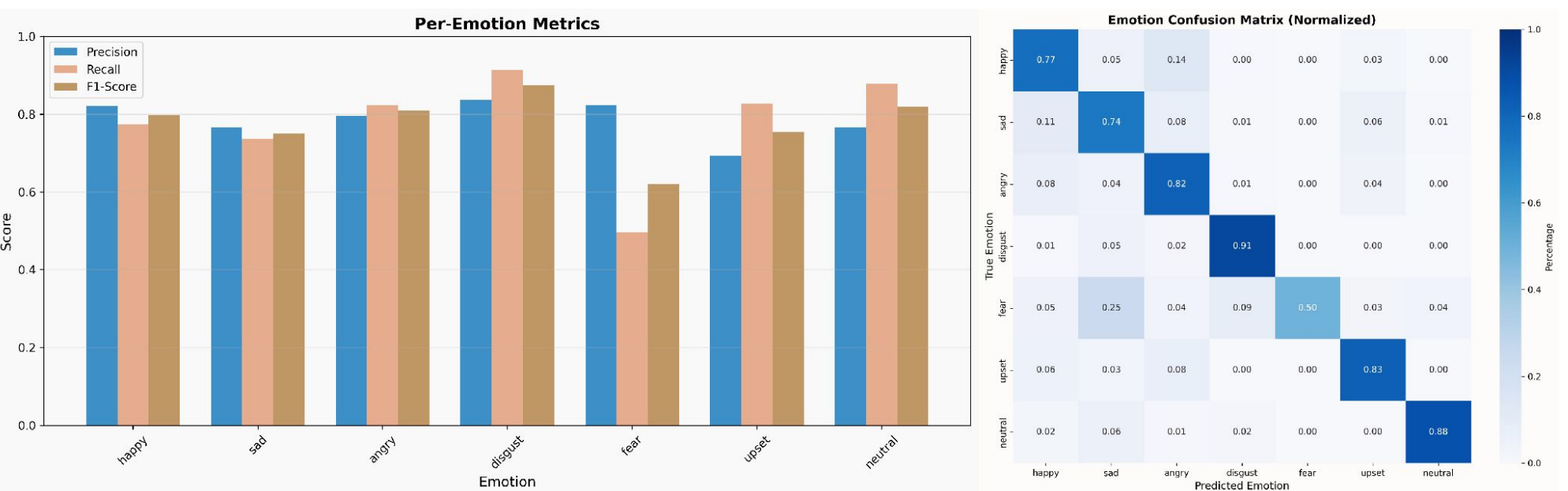


Fig. 3. **Left:** Per-emotion precision, recall, and F1-scores on the test set. Disgust achieves the highest F1 (87.4%), while fear remains the most challenging class (62.0%) due to data scarcity. Error bars denote 95% confidence intervals estimated via 1,000 bootstrap resamples. **Right:** Normalized confusion matrix on the test set. Values represent row-wise percentages (true emotion $\rightarrow$ predicted emotion). Diagonal entries indicate correct classifications. Major confusions occur between high-arousal emotions (happy $\leftrightarrow$ angry) and between fear and sadness.

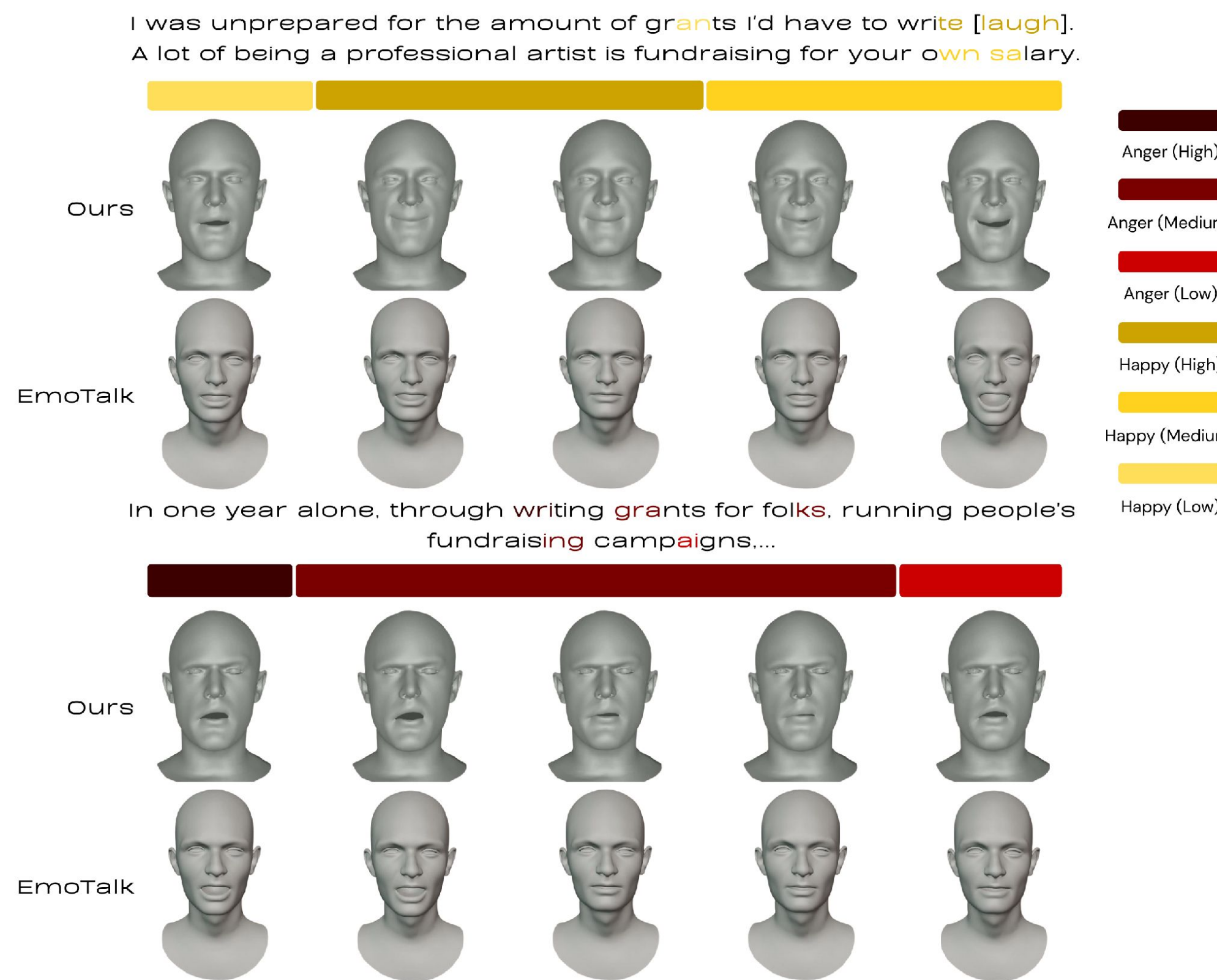


Fig. 4. Qualitative comparison of emotional facial animation between our method (Ours) and EmoTalk. The top example corresponds to an utterance predominantly expressing happy affect with varying intensity, while the bottom example is dominated by anger at different intensity levels. Colored bars indicate the predicted emotion category and intensity over time. Our method more faithfully follows the temporal dynamics of the speech, producing clearer transitions between low, medium, and high affect and more expressive, well-synchronized facial movements compared to EmoTalk.

LIMITATIONS & FUTURE WORK

- Fear remains the hardest class to recognize (F1 62.0%) due to data scarcity (2.0% of frames) and acoustic overlap with low-arousal states like sad.
- EmoVOCA pairs emotional 3D motion with neutral speech only; evaluation on genuinely emotional speech recordings remains future work.
- Non-standard EmoVOCA labels without clear acoustic correlates (e.g., "Drunk2", "Ill", "Moody", "Smile2", "Suspicious") were excluded; extending the taxonomy to cover these is an open direction.
- Broadening the corpus to more languages and accents beyond the current 4 English varieties would improve cross-lingual generalization.

APPLICATIONS

- Virtual assistants and digital avatars with emotionally expressive, real-time responses.
- Film and animation production and expressive dialogue-driven facial performance.
- Telepresence and virtual meetings with naturalistic emotional presence.



CONCLUSION

SEDTalker conditions 3D facial animation on temporally dense, frame-level emotion signals rather than one label per clip, enabling smooth, continuous emotional control without requiring paired emotional speech / 3D-mesh training data.

ACKNOWLEDGEMENTS & REFERENCES

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SCAN FOR CODE

